

Obstacle Detection, Depth Estimation And Warning System For Visually Impaired People

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Abstract—The work aims to develop a system which assists the blind people in sensing and identifying obstacles on their way. It also estimates the distance towards them. We use Yolov2 to train the system. With this, an image description is produced along with its distance from the person. The system would then present the person with an appropriate audio response. The range information is obtained by merging the output of object detector with that of an ultrasonic sensor mounted on an Auridino Uno. We evaluated the system both indoors as well as outdoors. The system can detect the object using the pre-trained deep neural network with 97% accuracy. We want to extend this work into a portable device like mobile phone by transfer learning of the model. The proposed system can identify more object categories along with depth estimation, with reasonable accuracy, while others can identify only the presence of an obstacle.

Index Terms—Assistive Technology, Visually Impaired, Depth Estimation, Object Recognition, Push Down Automata, YoloV2, Auridino Uno

I. INTRODUCTION

According to the Prevention of Blindness and Deafness Programme of WHO (World Health Organisation), about 285 million people are visually impaired worldwide. Thirty-nine million are blind, and 246 million have low vision (severe or moderate visual impairment). Among them, more than 15 million are from India. Blind people get information through other senses like hearing, taste, smell, and touch. The research in Assistive Technology is taking other sensory perceptions of the blind into account and help them to lead a normal life. Assistive Technology is any technology developed so as assist the living of a disabled person. Blind people learn about the environment through touch and voice input and output devices. There are haptic devices and voice recognition systems, text to the speech synthesiser which make use of these senses. The blind are not naturally endowed with the sixth sense of touch and hearing ability. But proper training on the above helps them to live independent like any other person with normal vision. In earlier days the blind people used the cane and dogs to guide them to navigate. The sensors are the motive force for the navigational devices. Sensors, together with many cameras, depth sensing, force or pressure sensors, enable the user to interact with the environment.

The following sections in our paper are organised as: Section II deals with related works, section III gives detailed design, implementation in section IV and results in section V followed by conclusion and references.

II. MOTIVATION AND RELATED WORKS

The average life span is increasing, but age-related muscular degeneration is a leading cause for visual impairment in humans. We aim to focus our work to detect objects and give a message to the blind regarding the obstacles in the path and the distance to them, and make them move around anywhere in the campus just like any other normal sighted person and thereby reduce their dependency on volunteers.

A. Object detection and recognition related works

Object detection and recognition is a well-studied problem in Computer Vision. Object detection deals with locating objects and object recognition deals with determining its height, range, the direction. There are many faster algorithms for object detection. The first real time face detection algorithm by Viola and Jones [1] used Haar basis and related complex filters to extract an integral image from the input image by some pixel operations. They also focus on selection of appropriate features from this and do a binary classification. The hand labelled upright face images were used for training. This has a wide range of applications in detecting objects like pedestrians and automobiles.

Another approach for object detection is by region proposal. In [2] Ren et al.(2017) introduces a region proposal network(RPN) which is a full convolution network which they found better than a selective search of Fast RCNN [3]. Fast RCNN together with two convolution layers, one for encoding position of conv map feature detector and second for objectness score and region proposal, forms a RPN. It takes in any image and produces rectangular object proposals, with object score by sliding the window over the convolution feature map or the conv map. They were able to do object detection using region proposal with accuracy 70.4% MaP (Mean Average Precision) on objects. They generate $n \times n$ spatial window of the input conv-feature map. Each sliding window is mapped to a lower dimensional vector; then data is fed into the box regression layer and a box classification layer. Then K-score is calculated for the value of each layer using logistic regression or softmax. For each box, Intersection over Union (IoU) is calculated. If IoU of a box is more significant than 0.7 that is assigned as a positive label, and if IoU is less than 0.3 then the box gets a negative label. Using softmax regression, we can find the optimal mergeable box so

that we can merge those boxes to find particular objects. A Fast RCNN [3] gives the entire image to a CNN(Convolution Neural Network) and produce feature maps. Fast RCNN uses selective search to find RoI which is time-consuming. Problem with Faster RCNN [2] is that, it requires a single image to undergo many passes to extract all images. Faster RCNN replaces this and is faster than the RPN.

Object recognition for 1000 different categories is done on a large CNN in [4] which consists of five convolution layers with an activation function ReLu, followed by max pooling layers. They introduced a drop out layer for a new regularisation method to reduce overfitting. Three fully connected layers and a 1000-way softmax at final dense layer. Humans labelled the images collected from the web using Crowd-sourcing tool by Amazon. This shows promising results of deep learning in object detection and recognition.

Real-time object detection is, even more, faster with Yolo [5] in which Redmon et al. (2016) divide the whole image into grids and identifies the objects in the bounding boxes. If a grid contains an object, calculate the confidence score of each grid concerning its position, height, width, and IoU(Intersection over Union). Then feed this into a convolution layer, and combine the grids with sliding window model. Using softmax regression we can find the optimal mergeable box so that we can merge those boxes to find particular objects. The given algorithm is fast, but it fails to capture the small details of the image like a flock of birds.

To achieve faster detection, Redmon et al.(2017) [6] used 1x1 convolution layers in the intermediate layers and few convolution layers in the top layers. They also used a fine grid to divide the images, to detect the small sizes of traffic signs.

B. Navigation related works

The precision and speed of object recognition improved rapidly with the use of CNN, and this can be applied to obstacle avoidance for the blind and visually impaired. An extensive survey on electronic travel aids for the visually impaired people is given in Dakapoulous et al. in [7]. A comparative study on the Electronic Travel Aid's usefulness and types of feedback by which the blind perceives is detailed in this paper.

It surveys the vision substitution mainly focusing on obstacle detection systems. The output is produced through tactile sense or auditory cues to indicate different directions regarding the obstacle, but results show that users can identify and discriminate objects in limited cases. Some of them were restricted to bulky prototypes while some were simple and portable. Dhrishti [8] was a prototype designed to assist the blind using the GPS to locate the blind.

Disparity based image processing is used to detect nearby objects in [9] by Paula et al. (2016). Though image analysis is not achieved in real time, fast implementation of Peano-Hilbert ensemble empirical mode decomposition is possible. Obstacles in a predefined route are detected, and the information is passed through the vibrotactile interface. Chen et al. [10]

(2018) incorporates 3D features, apart from the 2D features to solve the problem of shadows, occlusion and resolution in object detection by using the disparity map. A laser pointer and smartphone is used by template matching algorithm in [11]. The input video is processed, and acoustic feedback is given to the user. The usability of the system was poor as distance fluctuated.

Another approach for navigation is simultaneous localisation and mapping(SLAM) in robotic mapping. It means updating a map of an unknown environment while simultaneously keeping track of the agent locations with it. In [12], Zhang et al. (2017) sends the data from the camera to the server through WiFi, and the processing is done at the server, and the result is communicated to the user by speech. Cang et al(2018) in [13] proposed detecting objects using geometric contexts. In this, it segments 3D point cloud into planar objects and classifies each of them into a plane of a model object. It uses Gaussian mixture model to identify the model objects of one kind. It was able to detect doorways, staircase and a few non-structural objects in indoor while navigating.

Computationally efficient object detection discussed in the above section promises that evolution to perfection is possible with deep learning.

III. DESIGN

The proposed system given by Fig.1 consists of three modules.

- Obstacle detection with the help of neural networks
- Distance measurement with the help of sensors
- Generation of audio response for the finding of obstacle and distance

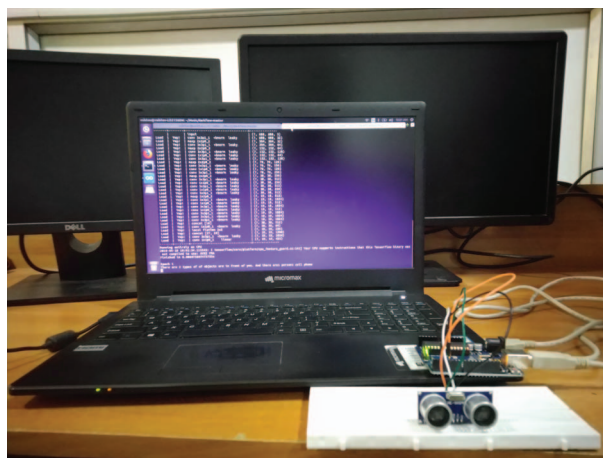


Fig. 1. The proposed system

A. Input

The input to our system will take snapshots from the webcam present in the laptop with 4GB RAM. Size of the image processed is 700*700 pixels. Along with the image, for distance measurement, we use ultrasonic sensor mounted upon an Arduino Uno.

B. Object Recognition

We are using a pre-trained neural network on COCO [14] dataset on our system. COCO dataset consists of images from complex daily scenes falling into 80 classes. The pre-trained model is trained with YOLO algorithm. The

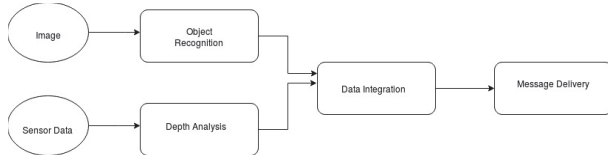


Fig. 2. Design Flowchart

advantage of using YOLO is that this model divides the image into grids of cells and each cell finds a confidence score of the individual grid as well as class probability. With the help of Bayes theorem, class detection score can be found by multiplying box confidence score with the class probability. YOLOv2 is faster as the grid size is smaller and more accurate than YOLO. Further, we can extend our training set to a wide range of objects in the real world.

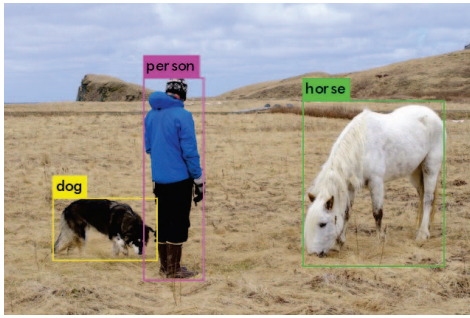


Fig. 3. A simple object detection which is done with the help of YOLOv2. Test image from [6]

C. Depth Analysis

The depth analysis is done with the help of the ultrasonic sensor. The sensor consists of two parts- a transmitter and a receiver. Figure 4 gives the basic working of the sensor.

- The transmitter transmits ultrasonic sound waves towards the object. The ultrasonic wave pulse is beyond the audible limit of human hearing. Transmission is done with the help of the transducer.
- The ultrasonic wave bounces back from the object body. The Receiver's job is to receive the reflected ultrasonic wave. The distance measurement with the help of the ultrasonic sensor is discussed in section IV.

Some of the advantages of using the ultrasonic sensor are:

- The detection is not affected by smoke, light, breeze etc. of the transmission medium.
- Ultrasonic wave can bounce back from most of the material's surface.

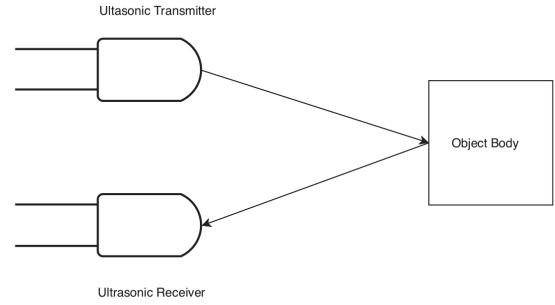


Fig. 4. Basic working of an ultrasonic sensor

D. Data Integration

We designed a merging module to integrate the results from both modules. This is discussed in detail in the implementation section. We are using a buffer to interact between Arduino and the object detection module. This module validates the data from the sensor and writes the data into a file and will be processed later for finding the distance.

E. Deliver the Message

The response generated in the previous stage will be displayed on the screen in simple text format, and this is for a partially blind man. For complete blind individuals, we are generating audio output with the help of gTTS (Google Text-to-Speech). gTTS is a Python library to interface with Google Translate text-to-speech API.

IV. IMPLEMENTATION

This project can be divided into three modules as follows:

- 1) Object Recognition
- 2) Depth Measuring
- 3) Audio Generation

The requirements and implementation details for each module are explained in the following sections

A. Object Recognition

Image prediction is done with the help of YOLOv2. It divides the image into small rectangles and generates a score for each rectangle using logistic regression. If the rectangle overlap score is more than 0.5, then we merge the rectangles. The network is trained on COCO dataset. The image frame is processed in real time with OpenCV library.

Image is processed to find the objects position. The algorithm returns the classified labels and location of the bounding box for the object. Process for finding nearest object detection from the image is given in Algorithm 1. Algorithm 1 takes as argument the processed image, an empty dictionary and an empty array. Our algorithm uses two functions $\text{dist}(a,b)$ and $\text{sort}(\text{array})$. $\text{dist}(a,b)$ returns the Euclidean distance from the point a to the point b and $\text{sort}(\text{array})$ returns the array in ascending order of the distances of objects from the blind.

B. Depth Measuring

Devices required for depth measurement are:

- Arduino Uno R3
- HC-SR04 Ultrasonic range finder

The module is fully integrated with the native operating system. This finds the distance between the obstacle and the person with the same principle as in figure 4. The sensor sends a burst of the ultrasonic wave towards the object and receives its echo after it bounces from the object surface. The system keeps track of time between sending and receiving the wave. With the help of time and speed of the wave through the medium, distance is calculated. The distance is calculated using following formula:

$$d = 1/2 * v * t \quad (1)$$

where d is the distance from sensor to the object, v is the velocity of the ultrasonic wave and t is the time taken between transmission and reception.

Algorithm 1 Finding the nearest object

Input: The Processed image

Output: Returns the nearest object from the photographer

dic = empty dictionary with three fields ($label$, $d1$, $d2$)

lis = empty array of type dic

l = width of the photo

h = height of the photo

$origin$ = tuple of(l , $h/2$)

for each element X of Result do

$dic[label] = X[label]$

pos = center of bounding box

$dic[d1] = dist(pos, origin)$

$dic[d1] = dist(pos, (pos[0], h/2))$

$lis.append(dic)$

end

sort(lis)

return $lis[0][label]$

C. Audio generation

For audio description of the image:

The output of object recognition is the input to this process. The Algorithm 2 for generating the description of the objects in the image the person takes as argument the processed image, the dictionary dic updated by Algorithm 1 and additional variables $no_classes$ and $no_objects$ which is the number of the distinct classes of objects present in the processed image and number of objects present in each class respectively. The generated 'str' variable is later fed into the gTTS module to convert into an audio file. The given string is converted into audio by connecting text-to-speech API of Google. The connection uses 'https' protocol.

Nearest object and its distance:

For finding the nearest object and the distance to it, we have implemented a PDA(Push Down Automata) given in figure 5.

Algorithm 2 Description of object in the image

Input: The Processed image and dic updated by Algorithm 1
Output: str which contains the description of the object present in the image.

$str = \text{"There are"} + no_classes + \text{"types of objects in front of you and there are"}$

for each element of dic do

$str = str + no_objects + \text{name of the class}$

end

A PDA is a finite state machine with a stack. A PDA is formally defined as a 7-tuple: $M = (Q, \Sigma, \Gamma, \delta, q_0, Z, F)$ where

Q = finite set of states

Σ = finite set of input alphabet

Γ = finite set of stack alphabet

δ = the transition relation

$q_0 \in Q$ = start state

$Z \in \Gamma$ = initial stack symbol

$F \subseteq Q$ = set of accepting states or final states

The attributes of the PDA are :

states = $\{1, 2, 2', 3, 4\}$

initial state = 1

final state = $\{3, 4\}$

input symbol = $\{Y, N\}$

output symbol = $\{a, b, c, d, e\}$.

Below are the stages and respective representation:

1 : find the nearest object from the output of IV.A

2 and 2' : correct sensor data

3 and 4 : final state

Here input symbol Y or N indicates the status of operation of that stage as successful or unsuccessful respectively.

Following are the outputs and respective representations:

a = "obstacle object-class" where object-class is class of the nearest object.

b = "is at X centimetre in front of you."

c = "please stop we are re-processing."

$d = \begin{cases} \text{"obstacle at X centimetre",} & \text{if } X \leq 15\text{cm} \\ \text{"please stop collision ahead",} & \text{otherwise} \end{cases}$

e = do nothing

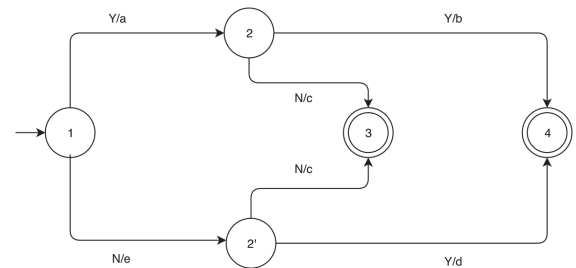


Fig. 5. PDA for nearest object with distance

The audio output for the input image in Figure 6 is: 'There is one person, one bottle and two TV monitors in front of you'

TABLE I
QUANTITATIVE EVALUATION OF OUR MODEL WITH RESPECT TO OTHER METHODS

Ref	Year	Sensor	A1	A2	A3	A4	A5	Demerits	Merits
[11]	2016	Laser Emitter	90%	Yes	No	No	No	Obstacle Color affected detection. Transparent objects not detected	Small objects less than 1cm can be detected with high resolution images.
[15]	2017	Ultrasound	-	-	No	Stairs	Yes	Heavy image processing time. US\$16 for GPS module	A wearable device prototype designed
[16]	2017	RGBD, Ultra-sound	98.5%	Yes	No	No	Yes	Object is not identified	Also find transparent obstacles
Proposed Method	2019	Ultrasound	97 %	Yes	Yes	80 types	Yes	Time for image description is more	Identify 80 classes of objects and produce auditory cues along with range.
A1: Accuracy (%), A2: Real Time Processing, A3:Object Identification, A4: Object Type, A5: Depth Estimation									

by Algorithm 2 and ‘the obstacle person is at 75 cms in front of you’, by the PDA.

TABLE II
EFFICIENCY OF OUR SYSTEM WITH RESPECT TO VARIOUS OBJECTS

Object	Object detection(%)	Accuracy of dist. result(%)
Dog	95	65
Person	97	85
Car	97	90
Bicycle	92	32
Bottle	89	20

V. RESULTS

We tested our system both indoors and outdoors. The performance of the system remains consistent. We also noted an anomaly in the ultrasonic sensor readings with increasing temperature. We ran our process for frames consisting of usual objects in indoor as well as outdoor. Quantitative analysis with respect to different other methods is discussed in Table I. The proposed system is able to identify more object categories along with depth estimation, with good accuracy while others identifies just the presence of any obstacle. The efficiency for different object categories is shown in Table II. The accuracy is calculated using the formula :

$$\eta = m/T * 100 \quad (2)$$

where m is the number of misclassifications and T is the total number of instances. Object detection accuracy is high. However, the accuracy of distance measurement was found varying for some objects. This was traced to the way sensor functions. It detects the distance from the bounced back ultrasonic wave. In case an object with low surface area is present in the frame, there exists a chance that most of the ultrasonic waves will not be incident on the object and that it will produce incorrect output. Figures 5 and 6 shows the recognised output for objects and the descriptions produced.

The time for executing one epoch purely depends upon the processing time and the length of the audio file generated, as



Fig. 6. Object detection which is done with the help of YOLOv2, detecting a person, bottle , and two monitors. The audio output is ‘The obstacle person is at 75cms in front you’. The description of the image is: ‘there is one person, one bottle and two monitors in front of you’.

we are trying to give constant feedback. The execution time for some instances is as shown in Table III. For the given scenario the average execution time is around 17.98 seconds per a single frame.

VI. CONCLUSION

Our work aims to build a cost-effective, simple, workable solution for visually impaired people. Our virtual assistant detects the object in real-time and generate responses regarding the nearest object along with its distance from the given person. This system for object detection works in real-time with Yolo algorithm and finds the nearest obstacle. The obstacle to blind range is measured with the help of ultrasonic sensor mounted upon an Arudino. The obstacle closer to the person need to be detected first for safe navigation. Using stereo vision cameras, the accuracy increases with distance and the cheap ultrasonic sensor gives a good accuracy. For

Table III : Execution time in seconds

Samples	Image processing	Image description	Nearest obstacle	Distance measurement	Total execution time
Image 1	3.303	9.215	0.00027585	7.563	20.082
Image 2	2.709	9.269	0.00029135	5.122	17.101
Image 3	2.711	10.327	0.00031567	5.099	18.137
Image 4	2.703	9.462	0.00031877	5.153	17.319
Image 5	2.750	9.115	0.00029182	5.436	17.301
Image 6	2.696	9.957	0.00017548	5.631	17.949
Average time	2.75	9.533	0.000028	5.440	17.976

future work, this solution can be extended to Android so that it can be used on a larger scale with interactive response delivery methods in real-time. Parallelism can be employed to optimise system utilisation and improve the response time.

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